

Гадиров Наиль
задание 17

1) Установить кросс-компилятор

```
gadirov@nail-gad:~/prog/kernel/linux$ ARCH=arm make CROSS_COMPILE=arm-linux-gnueabi- multi_v7_defconfig
HOSTCC scripts/basic/fixdep
HOSTCC scripts/kconfig/conf.o
HOSTCC scripts/kconfig/confdata.o
HOSTCC scripts/kconfig/expr.o
LEX scripts/kconfig/lexer.lex.c
YACC scripts/kconfig/parser.tab.[ch]
HOSTCC scripts/kconfig/lexer.lex.o
HOSTCC scripts/kconfig/menu.o
HOSTCC scripts/kconfig/parser.tab.o
HOSTCC scripts/kconfig/preprocess.o
HOSTCC scripts/kconfig/symbol.o
HOSTCC scripts/kconfig/util.o
HOSTLD scripts/kconfig/conf
#
# configuration written to .config
#
```

2) конфиг (arm)

```
gadirov@nail-gad:~/prog/kernel/linux$ ARCH=arm CROSS_COMPILE=arm-linux-gnueabi- make zImage
SYSHDR arch/arm/include/generated/uapi/asm/unistd-oabi.h
SYSHDR arch/arm/include/generated/uapi/asm/unistd-eabi.h
HOSTCC scripts/dtc/dtc.o
HOSTCC scripts/dtc/flattree.o
HOSTCC scripts/dtc/fstree.o
HOSTCC scripts/dtc/data.o
HOSTCC scripts/dtc/livetree.o
HOSTCC scripts/dtc/treesource.o
HOSTCC scripts/dtc/srcpos.o
HOSTCC scripts/dtc/checks.o
HOSTCC scripts/dtc/util.o
```

3) переносим в отдельную папку

```
gadirov@nail-gad:~/prog/kernel/linux/arch/arm/boot$ cp ../zImage ../../../../../.././Desktop/HW_17/
gadirov@nail-gad:~/prog/kernel/linux/arch/arm/boot$
```

4) собираем dtb

```
gadirov@nail-gad:~/prog/kernel/linux$ ARCH=arm make dtbs
SYNC    include/config/auto.conf.cmd
DTC     arch/arm/boot/dts/actions/owl-s500-cubieboard6.dtb
DTC     arch/arm/boot/dts/actions/owl-s500-guitar-bb-rev-b.dtb
DTC     arch/arm/boot/dts/actions/owl-s500-labrador-base-m.dtb
DTC     arch/arm/boot/dts/actions/owl-s500-roseapplepi.dtb
DTC     arch/arm/boot/dts/actions/owl-s500-sparky.dtb
DTC     arch/arm/boot/dts/airoha/en7523-evb.dtb
DTC     arch/arm/boot/dts/allwinner/sun4i-a10-a1000.dtb
```

5) отправляем к нам

```
vexpress-v2p-ca15-cc1.dtb vexpress-v2p-ca9.dts
gadirov@nail-gad:~/prog/kernel/linux$ cp arch/arm/boot/dts/arm/vexpress-v2p-ca9.
dtb ../../../../Desktop/HW_17/
gadirov@nail-gad:~/prog/kernel/linux$
```

6) запускаем через qemu

```
gadirov@nail-gad:~/Desktop/HW_17$ QEMU_AUDIO_DRV=none qemu-system-arm -M vexpres
s-a9 -kernel zImage -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0" -nographic
[    0.000000] Booting Linux on physical CPU 0x0
[    0.000000] Linux version 7.1.0-07602-g66affa37cfac (gadirov@nail-gad) (arm-l
inux-gnueabi-hf-gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0, GNU ld (GNU Binutils
for Ubuntu) 2.42) #1 SMP Thu Jun 18 03:37:01 +07 2026
```

7) падаем

```
[ 1.636064] 0101          65536 ram15
[ 1.636069] (driver?)
[ 1.636090] 1f00          131072 mtdblock0
[ 1.636099] (driver?)
[ 1.636145] List of all bdev filesystems:
[ 1.636161] ext3
[ 1.636170] ext2
[ 1.636179] ext4
[ 1.636188] squashfs
[ 1.636196] vfat
[ 1.636205] msdos
[ 1.636213] ntfs
[ 1.636224]
[ 1.636357] Kernel panic - not syncing: VFS: Unable to mount root fs on unknow
wn-block(0,0)
[ 1.638887] CPU: 0 UID: 0 PID: 1 Comm: swapper/0 Tainted: G      W
7.1.0-07602-g66affa37cfac #1 VOLUNTARY
[ 1.639015] Tainted: [W]=WARN
[ 1.639049] Hardware name: ARM-Versatile Express
[ 1.639106] Call trace:
[ 1.639136] unwind_backtrace from show_stack+0x10/0x14
[ 1.639248] show_stack from dump_stack_lvl+0x54/0x68
[ 1.639321] dump_stack_lvl from vpanic+0x234/0x454
[ 1.639386] vpanic from do_panic_on_target_cpu+0x0/0x14
[ 1.640007] ---[ end Kernel panic - not syncing: VFS: Unable to mount root fs
on unknown-block(0,0) ]---
```